Instrumentation - 2: A Practical Handout for Olympiad

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Contents

1 Telescope Mounts **3**

1.1 Basic Altazimuth Mount . . . 3

1.1.1 Manual Handling . . . 4

1.1.2 Field Rotation: Avoid Alt-az Mount for Astrophotography . . . 5

1.2 Basic Equatorial Mount (EQ) . . . 6

1.2.1 Components in EQ mount . . . 7

1.2.2 Polar Alignment . . . 12

1.3 Dobsonian Mount (Altazimuth) . . . 13

1.3.1 Design . . . 14

1.4 German Equatorial Mount (GEM) . . . 15

1.4.1 The Meridian Flip . . . 15

1.5 Fork Mount . . . 16

1.6 Horseshoe Mount . . . 17

2 Accessories **18**

2.1 Finderscope . . . 18

2.1.1 Aligning the Finderscope . . . 19

2.2 Eyepiece . . . 20

2.2.1 Eyepiece Internal Ray Diagram . . . 22

2.2.2 Field Stop Diameter . . . 23

2.2.3 Exit Pupil . . . 23

2.2.4 Eye Relief . . . 23

2.3 Barlow Lens . . . 25

2.3.1 How Barlow Works . . . 26

2.4 Focuser . . . 26

2.5 Filters . . . 27

2.5.1 Broadband Filters . . . 27

2.5.2 Narrowband Filters . . . 28

2.5.3 Solar Filter . . . 29

2.5.4 Light Pollution Filters . . . 29

2.6 Stopwatch . . . 29

3 Further Resources **32**

Prerequisite

Before starting Instrumentation - 2, it's an absolute necessity that you understand stuffs form Instrumentation - 1. You can find it here. Also, one need to have a fundamental grasp of positional astronomy and coordinate systems. You can follow AO Guide - Positional Astronomy. If you don't have a telescope to play around with, watch videos on the internet to familiarize yourself.

1 Telescope Mounts

A telescope mount is the mechanical platform that holds and moves the optical tube assembly. Its primary purpose is to provide a stable, movable support structure that allows the telescope to be pointed at celestial targets.

Mounts are classified by their axes of rotation and tracking method. The two fundamental types are the altazimuth mount, which moves in altitude (up/down) and azimuth (left/right), and the equatorial mount, which aligns one axis with Earth's rotational axis (Right ascension α and Declination δ).

Olympiad relevancy:
In the IOAA and the IAO, the telescope round often tests mount operation skills. Different mounts appear each year: a German equatorial mount was used in IOAA 2025 and IAO 2023, a Galileoscope with its simple altazimuth mount in IOAA 2024, and a Dobsonian mount in IOAA 2023. We'll mostly talk about an Alt-az, a German equatorial, and a Dobsonian. There are some other mounts, like a Fork mount or Horseshoe mount, but they aren't often seen in IOAA.

1.1 Basic Altazimuth Mount

The altazimuth mount is a two-axis mount that allows movement in altitude (vertical angle) and azimuth (horizontal angle). Its name is derived from "altitude" and "azimuth." The altitude axis (an imaginary line about which a body rotates) is horizontal and allows the telescope to tilt up and down. The azimuth axis is vertical and allows the telescope to rotate left and right parallel to the ground plane. This design mimics the natural motion of looking around the sky: up/down and left/right.



A basic altazimuth mount consists of a base that rotates on a bearing around the vertical azimuth axis. Mounted on this base is a cradle or fork that holds the telescope tube and pivots around the horizontal altitude axis. The altitude axis must be precisely perpendicular to the azimuth axis to avoid pointing

errors. Friction, bearing quality, and balance determine the smoothness of motion.

To track an object automatically, an altazimuth mount requires two independent motors,one on each axis,driven at continuously variable, non-constant rates. The required angular velocities $\dot{A}$ (azimuth rate) and $\dot{a}$ (altitude rate) for an object at azimuth A, altitude a, latitude ϕ, and sidereal rate ω are given:

$$\dot{A} = \omega(\sin\phi + \cos\phi\cos A\tan a)$$
$$\dot{a} = \omega\cos\phi\sin A$$

1.1.1 Manual Handling

The manual movement is typically controlled through manual slow-motion controls and free-hand motion.

- **Free-hand movement**
  Free-hand movement refers to directly grasping the telescope tube or a handle on the mount and physically moving it in altitude and azimuth. This is common on simpler mounts, especially Dobsonians and basic tripod-mounted refractors. The telescope tube is typically held in place (locked at one exact alt-azimuth) by adjustable friction clutches or locking knobs. When these clutches are loosened, the telescope can be moved smoothly and freely in any direction. This freehand movement is used for initial rough pointing toward a target.
- **Fine-adjustment cables**
  Once the target is approximately centered in the finderscope or eyepiece, the observer tightens the clutches just enough to allow fine adjustment. Most manual altazimuth mounts are equipped with slow-motion control knobs or fine-adjustment cables. These are connected to worm gears or tangent arms that provide small angular movements without disturbing the telescope's balance.

  The altitude slow-motion knob moves the telescope up and down, while the azimuth slow-motion knob rotates it left and right. You can use these knobs to move the telescope at a controlled, slow rate, for accurate centering of objects in the field of view. These controls are necessary for tracking objects manually during observation, and the observer must continuously adjust both axes to compensate for Earth's rotation.



Above is how a fire-adjustment cable looks (enclosed in a yellow shape, kinda contrasts with the white background, but it's fine). In observational tasks during astronomy olympiads, participants may be required to use these knobs to point at the celestial targets. It's better to have hands-on experience with the telescope.

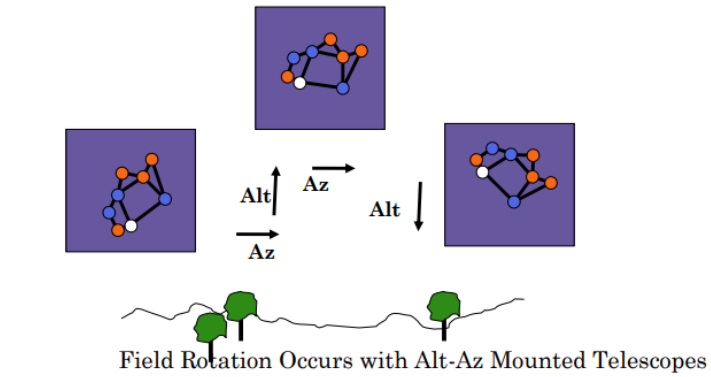
Task 1: IOAA 2024

Check how the telescope round was in IOAA 2024. If you have a telescope, try to mimic something similar. here is the question (Page 5-7), and here is the solution (Page 9-14).

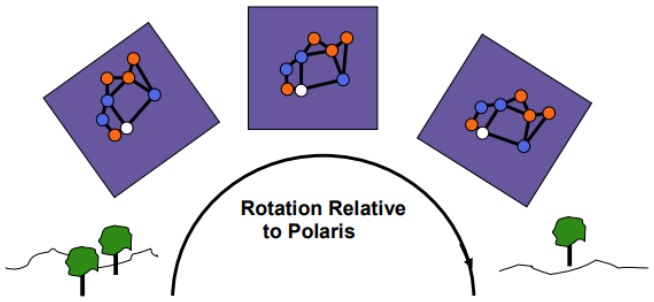
1.1.2 Field Rotation: Avoid Alt-az Mount for Astrophotography

This content is from StarMaps101 by Fahim Rajit Hossein.

Alt-az mounts are to be avoided due to a phenomenon known as field rotation. Field rotation occurs due to the design of the alt-az mount. One axis, the azimuth axis, rotates around a vertical line passing from the center of the telescope to the zenith (left to right). The altitude axis rotates about a horizontal axis (up and down). Both axes must rotate at different rates to maintain the object in the field of view (FOV). As the Earth rotates about its axis, the central object in the field of view will remain centered, but other objects in the FOV will appear to rotate with elapsed time. The orientation of the sky in your FOV slowly twists, and overtime star forms arcs instead of points, which eventually ruins long exposure. Field Rotation is an undesirable effect on time-exposure astrophotographs caused when an "Alt-Az" ("altitude-azimuth" or alt-azimuth) mount is used to hold and point a telescope with a camera. It is also sometimes called Frame Rotation because the objects framed by the photograph seem to have rotated while the picture was taken. Check this video here



Field Rotation Occurs with Alt-Az Mounted Telescopes



Zero Field Rotation with Equatorial Mounted Telescopes

Let the FOV of the telescope be represented by a square and follow the motion of Sagittarius as it rises in the east. Because the alt-az mount can only move around the azimuth axis and up and down on the altitude axis, the FOV remains fixed with the bottom parallel to the horizon. The objects in the sky, however, are rotating around the celestial pole star, Polaris. So asterisms and binary stars appear to rotate in the FOV. An equatorial mounted telescope does not suffer from field rotation because the

telescope is polar aligned, and the FOV does not remain horizontal as it tracks; the telescope rotates. So the FOV matches the rotation of the object, and it remains in the same fixed position.

Mathematics behind Field Rotation:
Stars appear to rise in the east and set in the west, rotating about the north celestial pole star, Polaris. Rising stars will reach their highest elevation in the sky as they cross the local meridian, due south of the line between Polaris and the zenith. The altitude indicates the vertical position of the star with 90° at the horizon and 90° at the zenith. From direct observation, the following aspects are known about the rate of rotation:

1. The maximum rate of field rotation occurs as the star passes through the zenith.
2. The rate of field rotation is minimal when a star passes through the prime vertical at the horizon.
3. Considering the azimuth only, the rate of field rotation is highest when the star passes through 0° (N) and 180° (S), and lowest when the star passes through 90° (E) and 270° (W).
4. Regarding latitude, field rotation is highest at the Earth's equator and non-existent at the NCP and SCP.

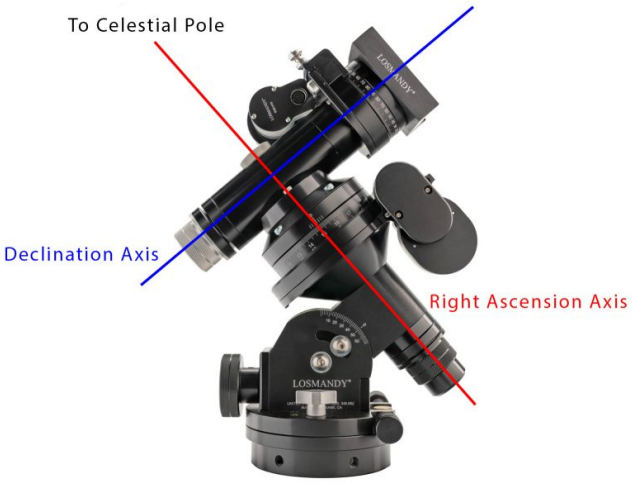
The field rotation rate $\dot{\theta}$ (in degrees per second) for an observer at latitude ϕ, tracking an object at altitude a and azimuth A, with sidereal angular velocity ω, is given by:

$$\dot{\theta} = \frac{\omega \cdot \cos\phi \cdot \cos A}{\cos a}$$

where, $\omega = 4.178 \times 10^{-3}$ degrees/second; Earth's rotation. For small altitudes, $\cos a \approx 1$, and the rotation is approximately constant. Near the zenith ($a \to 90°$), $\cos a \to 0$, making the rotation rate theoretically infinite,this is the "zenith blind spot" where altazimuth tracking fails.

1.2 Basic Equatorial Mount (EQ)

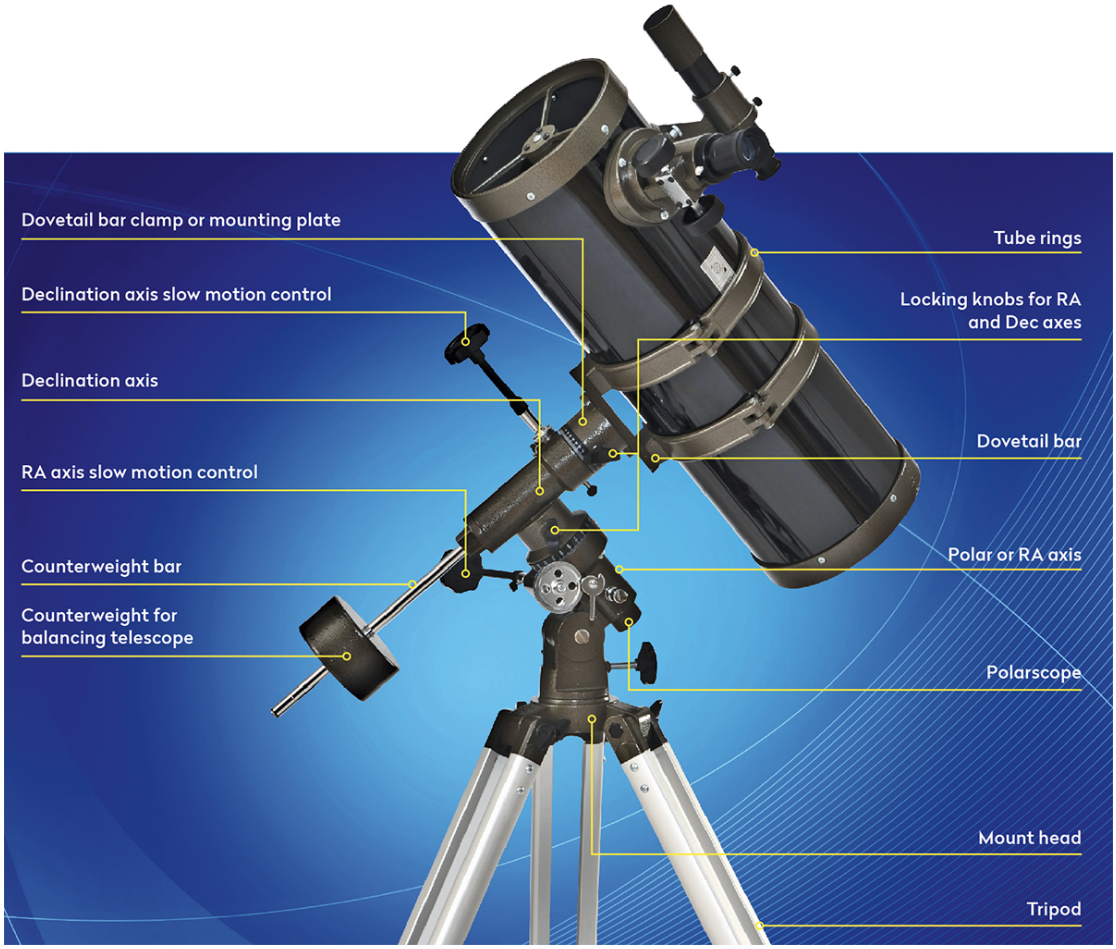
The equatorial mount is a telescope mounting system designed with one rotational axis aligned parallel to Earth's rotational axis. This axis is called the polar axis or right ascension axis. The second axis, called the declination axis, is perpendicular to the polar axis. When properly aligned, the polar axis points toward the celestial pole. Check this YouTube video here.



Why EQ mount over Altazimuth Mount?
Equatorial mounts are preferred over alt-az mounts because they require motion in only one axis to

track (during long exposure astrophotography. It follows the rotation of sky, so its easier to track the object. They also eliminate field rotation during long-exposure astrophotography. Additionally, their axes align directly with celestial coordinates, which makes object location and navigation more easier for observational astronomy.

1.2.1 Components in EQ mount



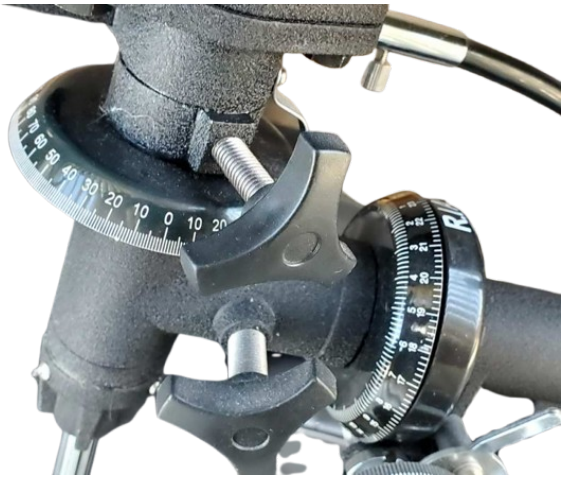
1. **Polar or Right Ascension (α) axis:**
   The Right Ascension (RA) axis is the primary rotational axis of an equatorial mount. It is aligned parallel to Earth's rotational axis, pointing toward the celestial pole. When properly aligned, this axis remains fixed relative to the stars while Earth rotates beneath it.

   The RA axis has sidereal tracking. During astrophotography, by rotating the telescope around this axis at exactly $15.041''$ per second (in a motorized system) (one complete revolution in 23 hours 56 minutes 4 seconds), the mount compensates for Earth's rotation. This constant tracking rate applies to all celestial objects regardless of their position in the sky. In equatorial coordinates, rotation around the RA axis changes only the right ascension coordinate of the telescope's pointing direction. The declination remains constant during pure RA motion.

2. **Declination (δ) axis:**
   The Declination (Dec) axis is perpendicular to the RA axis. Rotation around this axis changes only the declination coordinate of the telescope's pointing direction, while right ascension remains constant during pure declination motion.

During astrophotography, the declination axis is fixed relative to the RA axis. This means that once an object is centered in the field of view, only the RA axis needs to move to track it. The declination axis remains stationary during tracking, except when making corrections for polar alignment errors or guiding.

3. **Setting Circles (Dial)**
   Equatorial mounts often feature graduated dials called setting circles on both the Right Ascension (RA) and Declination axes. These circles allow the observer to set or read the telescope's pointing coordinates directly without electronic aids.



   (a) **Right Ascension Setting Circle**
   The RA setting circle is mounted on the polar (RA) axis. It is typically marked in hours, minutes, and sometimes seconds, covering the full 24-hour range of right ascension from 0h to 23h 59m 59s. However, since sidereal time advances at a rate slightly different from solar time, the RA circle corresponds to sidereal hours. To use the RA circle effectively, the observer must first synchronize it with the current sidereal time or the RA of a known star. The procedure is:

   1. Polar align your telescope. (*We'll deal with it in the next subsection*)
   2. Point the telescope at a star of known right ascension.
   3. Rotate the RA setting circle until it reads the star's RA value.
   4. The circle is now calibrated; pointing to other objects involves rotating the telescope until the circle reads the target's RA.

   Example: Suppose you want the setting circles calibrated. This is done by pointing the telescope at a bright reference star of known coordinates, such as Vega (RA 18h 37m, Dec +38° 47′). The observer centers Vega in the eyepiece, then rotates the RA circle until it reads Vega's RA (18h 37m), and adjusts the Dec circle to read Vega's Dec (+38° 47′).

   (b) **Declination Setting Circle**
   The Declination setting circle is mounted on the declination axis. It is marked in degrees from −90° to +90°, corresponding to the full range of declination coordinates. The declination circle is fixed relative to the telescope tube and indicates the current declination pointing. Unlike the RA circle, it does not need frequent recalibration once properly zeroed. The circle can be zeroed using a star of known declination.

Setting circles provide a manual method for locating celestial objects by coordinates. First, you need to calibrate both circles using a known reference star. Then, for a target at coordinates (α, δ), rotate

the telescope in declination until the Dec circle reads δ, and right ascension until the RA circle reads α.

Idea 1: Intuitive understanding

Imagine you're at an astronomy camp with a manual equatorial mount telescope. You polar-align it perfectly, then point it at Vega (RA 18h 36m) at exactly 7:00 PM local time (which is at your meridian), and you even set the RA setting circle to 18h 36m. Then you just... leave it there, no tracking, no touching. You come back at 3:00 AM to show your friend Vega, but it's totally not in the eyepiece anymore, and you only see some random patch of sky. Your friend asks, "Hey, what RA does the setting circle show right now, and why isn't Vega there?" Explain what went wrong and calculate roughly how many hours of RA the circle is off by now, assuming Sideral Time = Solar Time.

Answer:
Earth is rotating. So even though they set the circle to Vega's RA at 7 PM, the mount isn't moving, so the sky basically slides past the telescope. Vega drifted west out of the field, minutes after they left.

When they come back at 3 AM, that's 8 hours later. The setting circle is still stuck at 18h 36m because they never moved the scope, but the sky's meridian has moved forward in RA by about 8 hours (since 1 solar hour $\approx$ 1 sidereal hour roughly). So the circle is literally 8 hours behind , it shows 18h 36m, but the meridian is now at like 2h 36m RA (next day). That's why Vega isn't there; it set hours ago, and they're pointing at some random RA from earlier. So, if they don't track, their RA setting circle becomes useless after a short time. You again have to recalibrate it again as mentioned above.

Example 1: Point to a star

Consider the star Antares, which has approximate coordinates:

$$\text{Right Ascension (RA)} = 16\text{h } 29\text{m } 24\text{s}, \quad \text{Declination (Dec)} = -26^\circ 25' 55''$$

To point at Antares, the observer rotates the telescope in declination until the Dec circle reads approximately $-26^\circ 26'$. Then, while keeping the declination fixed, the observer rotates the telescope along the RA axis until the RA circle reads 16h 29m 24s. The telescope should now be pointed at Antares.

Task 2: IOAA 2015

Check this IOAA 2015 (Main Telescope) Page 1, here. Familiarize yourself with these kinda Observational tasks. We'll talk about some in detail as we go further.

4. **Manual Handling of Equatorial Mounts**
Equatorial mounts include manual controls for both coarse and fine positioning. These controls allow the observer to point the telescope without motorized assistance.

   (a) **Locking Knobs and Free Movement**
   Each axis of an equatorial mount is equipped with a locking knob or clutch mechanism (check the figure on page 7). When these knobs are loosened, the telescope can be moved freely by hand

along that axis. This allows for rapid, large-angle movements. For example, to move the telescope from Right Ascension 6h to 14h, the observer would:

i. Loosen the RA locking knob.

ii. Physically (with free hand) rotate the telescope tube around the RA axis until it is approximately at the desired position.

iii. Tighten the RA locking knob to secure the position.

The same procedure applies to the declination axis: loosening the Dec knob allows the telescope to be tilted up or down freely. Once the target is roughly centered in the finderscope, the knob is tightened to prevent further movement

(b) **Fine Adjustment Cables (Slow-Motion Controls)**
After locking the axes approximately on target, fine adjustment is done using slow-motion control cables (check the figure on page 7). These are flexible cables connected to gears on each axis. Turning the cable's knob rotates the gear, moving the telescope slowly.

The RA slow-motion control rotates the telescope in right ascension at a rate much slower than manual turning. One full turn of the knob might move the telescope only a few arcminutes in RA. The Dec slow-motion control provides similar fine movement in declination. It is used to center objects vertically in the field of view. Unlike the RA control, it does not need to be continuously turned during tracking, unless correcting for polar alignment errors.

Tip 1: Handling in IOAA 2025 (My Personal Experience)

So in IOAA 2025, the telescope exam was indoors, we had to point the scope at a screen showing a star field and match it to RA and Dec coordinates we'd calculated earlier. The catch was that the equatorial mount's RA axis was locked, so you could only move it using the fine adjustment knob on the polar axis. The Dec axis, though, was completely free,you could just tilt the tube by hand.

Took me a minute to figure that out, ngl. I wasted like two minutes trying to use the fine adjustment for declination because that's what we practiced with in camp, and I thought that knob was basically detached or broken in the exam setup. Well, I was panicking so much that it eventually ruined that part of the test (OT 01.3). Once I realized it, I just moved the declination axis with freehand (without the knob). So, in the olympiad test, always check which axes are actually drivable before you start. Don't assume all knobs work. Sometimes you just gotta freehand it.

Also, don't forget to remove the cover from the telescope aperture and the eyepiece :D (From Trishav's (Nepal's IAO and IOAA delegate) and Teodor's (Romanian IOAA, IOAA Jr & IAO Medalist) IAO 2023 experience).

You can check the question here. and here are the things we saw on the screen.

5. **Dovetail mount**
Alright, so about dovetails,they're basically the metal rails that connect your telescope tube to the mount head. They slide into a clamp on the mount and get locked in with one or two screws. Super simple, but important. (In the photo below, Lovemandy and Vixen are the names of the producers, not a specific type of dovetail). And also, check the figure on page 7, that reflector has its own tube ring, and you can slide that in the dovetail. Check any of these YouTube video from here.



Tip 2: Accidental slipup in IOAA

In the heat of an olympiad round under time pressure and all, it's easy to confuse the dovetail locking screw with the RA/Dec locking knobs.

- The dovetail lock just holds the telescope on the mount. If you loosen it mid-task, your whole scope can slide out or rotate. Tighten it once at the start and leave it alone.
- The RA and Dec locks control whether you can move the telescope around each axis. Loosen them to move freely, tighten to hold position.

If you accidentally loosen the dovetail instead of an axis lock, your tube might swing or droop, and you'll waste time re-balancing. Before you even touch the fine-adjustment knobs check if the dovetail is secure or not (Give the tube a gentle wiggle). Also check, which axis locks are actually working? (RA locked? Dec free? Vice versa?)

6. **Latitude Adjustment**

The latitude adjustment on an equatorial mount sets the angle of the polar axis to match your observing location's latitude. Basically, you tilt the mount so it points at the celestial pole,Polaris in the north, Sigma Octantis in the south.

There's usually a scale on the mount side with degrees marked. You set it to your latitude (like 27° for Kathmandu, 40° for New York). Some mounts have a locking bolt; others use a fine-adjust screw. Once set, you don't need to touch it unless you move to a different location.

Tip 3: Latitude adjustment

Say in IOAA, you're asked to point to a certain (α, δ) (like in IOAA 2015). If you don't set the latitude right, polar alignment will be off from the start and eventually will drift in Declination, even if you try to align manually. In timed tasks, you might waste minutes trying to fix drift instead of doing the actual pointing.

Check the latitude scale before you start. If it's way off, fix it first. During indoor rounds, they usually pre-set it, but verify anyway. If the mount has a fine-adjust for latitude, use it for final tweaking after rough polar alignment.

7. **Counterweight System in Equatorial Mounts**

The counterweight is a weighted mass mounted on a shaft opposite the telescope on German equatorial

mounts. Its purpose is to balance the telescope's weight around the polar (RA) axis. (check the figure on page 7)

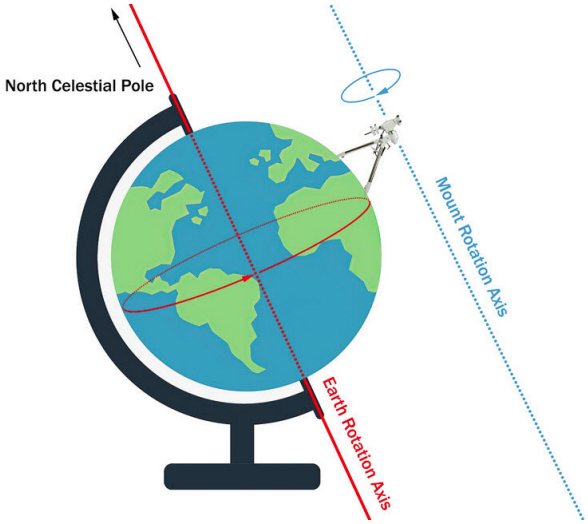
The counterweight shaft extends from the opposite side of the polar axis from the telescope. The counterweight itself slides along this shaft and can be locked at different positions. The basic principle is torque balance:

$$W_{\text{scope}} \times d_{\text{scope}} = W_{\text{cw}} \times d_{\text{cw}}$$

Where W_{scope} is the telescope weight, d_{scope} is its distance from the RA axis, W_{cw} is the counterweight mass, and d_{cw} is its distance from the RA axis. Check this YouTube video here. And, to balance the Declination axis, adjust the telescope's position in its mounting rings (tu until it remains balanced at any Dec angle. It's just an iterative process.

1.2.2 Polar Alignment

Polar alignment means setting your mount's polar axis to point exactly at the celestial pole (North or South). This aligns the mount's rotation axis with Earth's, so you only need to turn one knob (RA) to track stars at a constant sidereal rate, because the declination axis stays fixed relative to the star's position. Without accurate polar alignment, objects will drift out of view even if you're tracking, and the drift will be in both RA and Dec directions, making long-exposure astrophotography impossible. Once aligned, setting circles becomes accurate, and manual tracking with slow-motion controls works smoothly along a single axis. In IOAA 2017 & 2019, the telescope had come with polar alignment by defauly. You didn't have to care for that :D



Setup:
First, set the mount's latitude scale to your location's latitude. Then point the polar axis roughly north (south in Southern Hemisphere). Use a compass, but remember magnetic declination,true north isn't the same as magnetic north. You could use Stellarium to locate Polaris (Not applicable for IOAA tho). And at night, you can just eyeball it toward Polaris if you're in the north.

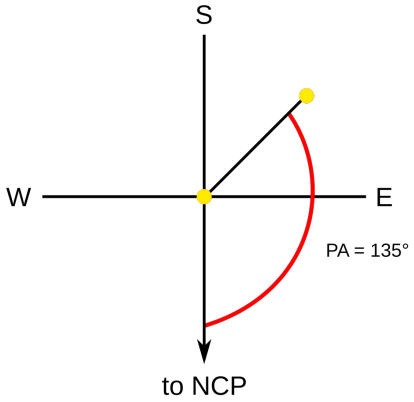
Polar Alignment Using Polaris (Northern Hemisphere):
This method uses Polaris (α Ursae Minoris) to align the equatorial mount's polar axis with the celestial north pole. Polaris is not exactly at the pole; it orbits around it at a radius of approximately $0.75°$ ($45'$). For IOAA, if the test is outdoor, then this shall work as they won't ask for long exposure astrophotography

stuff. But, in real life, this error would eventually pileup causing an error in declination. Watch this YouTube video here. You can you Polarscope method too, which is very accurate. Watch this YouTube video to see how a polarscope is used here.

Position Angle (Precise positioning of the Celestial Pole)

If you don't have a polar scope, you can still accurately align using Polaris by knowing its position angle. It is the direction from the true celestial north pole to Polaris, measured east from north. Think of it like a clock face centered on the true pole, with Polaris moving around it every 24 hours.

Polaris orbits the true pole at a fixed distance of about $0.75°$ ($45'$). The position angle (PA) tells you where along that circle Polaris is at any given time. It's measured in degrees.



- $0°$ = Polaris directly above the pole (north)
- $90°$ = Polaris to the right of the pole (east)
- $180°$ = directly below (south)
- $270°$ = to the left (west)

For any star (α, δ), Local sideral time being (LST) and observer being at latitude ϕ, the position angle is given by:

$$\tan(\text{PA}) = \frac{\sin(\text{LST} - \alpha)}{\tan\phi \cos\delta - \sin\delta \cos(\text{LST} - \alpha)}$$

For, Polaris' coordinates (α_{Pol}, δ_{Pol}), Since Polaris' declination is very close to $90°$, $\cos\delta_{Pol} \approx 1$ and $\sin\delta_{\text{pol}} \approx 1$, so a simpler approximation works:

$$PA \approx \alpha_{\text{pol}} - LST$$

Example: If LST = 8h and $\alpha_{\text{Pol}} \approx 2.5$h, then PA $\approx (2.5 - 8) \times 15 = -82.5°$. So Polaris is about $360° - 82.5° = 277.5°$ east of north relative to the true pole.

Now, you need to mentally picture the relationship between the true celestial north pole and Polaris. The position angle (PA) tells you exactly where Polaris sits around the pole at that moment. Now you need to adjust the mount's polar axis so that Polaris is positioned at this calculated offset from where the axis is pointing.

1.3 Dobsonian Mount (Altazimuth)

The Dobsonian mount is a specialized altazimuth mounting system designed for Newtonian reflectors. It's characterized by its simplicity, stability, and low cost per inch of aperture. It was developed by John Dobson in the 1960s, and it revolutionized amateur astronomy by making large-aperture telescopes accessible. The mount consists of two main components: a circular ground board typically made from plywood or similar material, and a rocker box. The telescope tube is pivoted in the rocker box. Watch this YouTube video from here.

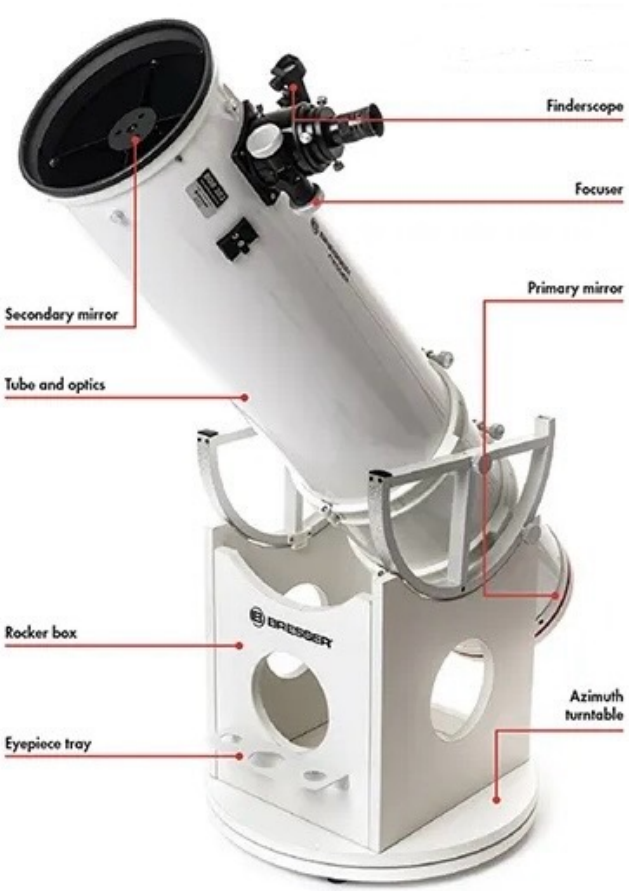


Why Use it?

So, for everyday amateur astronomy, a Dobsonian is basically the GOAT for beginners. It's basically a super stable alt-az mount made of cheap materials like plywood and Teflon, so you get a huge aperture for way less money than an

EQ mount. It's stupid easy to use,just point and look,without the annoying setup and counterweights of an EQ mount. You get significantly more light-gathering power for your buck than a small, wobbly alt-az mount or an expensive, complicated EQ mount in the same price range.

1.3.1 Design



1. **Azimuth Bearing**
   The azimuth bearing consists of a flat, circular ground board. Three Teflon pads are mounted on this board in an equilateral triangle configuration. The rocker box rests on these pads, with its bottom surface made from a smooth, low-friction material. The azimuth bearing allows the entire telescope to rotate horizontally (360°).

2. **Altitude Bearing**
   The altitude bearing consists of large, circular bearing surfaces attached to the sides of the telescope tube. These bear against pads mounted on the sides of the rocker box. The altitude bearing allows the tube to tilt vertically.

3. **Manual Movement**
   Tracking an object requires manual movement in both axes simultaneously. As Earth rotates, the observer must gently push the telescope to follow the object's motion. This is done by applying light pressure to the tube. The bearing surface is smooth, and it allows precise, small movements. There are no gears, motors, or slow-motion controls (fine adjustment cables) in a basic Dobsonian. So, tracking is entirely manual using your hands. The required tracking rates depend on the object's altitude and

azimuth, which is the same as the altaz mount.

$$\dot{A} = \omega(\sin\phi + \cos\phi\cos A\tan a)$$
$$\dot{a} = \omega\cos\phi\sin A$$

where $\dot{A}$ is azimuth rate and $\dot{a}$ is altitude rate for an object at azimuth A, altitude a, latitude ϕ, and sidereal rate ω.

4. **Balance**
   For balance, the telescope tube must be positioned so its center of gravity aligns with the altitude bearing axis. If the tube is too front-heavy, it will tend to tilt downward. If too back-heavy, it will tilt upward. Balance is adjusted by sliding the tube forward or backward in the rocker box, or by adding small counterweights.

Task 3: IOAA 2023

The Telescope test was indoors in IOAA 2023. They used a Dobsonian inside a Planetarium. You had 30 minutes to do actual observations of simulated screens showing an asteroid occultation and Starlink satellites, then another 30 to write everything up. For the asteroid, you had to time exactly when it blocked a star and for how long. With the satellites, you measured how fast they moved across the sky and calculated how far apart they were in space. Then there was a planet image where you had to identify five of its moons, and finally, you had to find and measure a new "supernova" in a galaxy image. Check the problems from here

1.4 German Equatorial Mount (GEM)



German equatorial mount (GEM) is basically the classic EQ mount you've probably seen pictures of. It has the same two axes as any equatorial mount: the polar axis (RA) and the declination axis (Dec). The polar axis points at the celestial pole, and you track by just rotating that axis at the sidereal rate. The Dec axis lets you move north/south in the sky. The working mechanism is the same as we talked in 1.2 Basic Equatorial Mount (EQ), IT'S NO DIFFERENT! In this section, I'll talk about the specifics of a GEM.

You'll mostly find a GEM in IOAA. What makes the GEM special is its T-shaped design. The telescope sits on one end of the Dec axis, and a counterweight sits on the other end to balance it. The whole thing rotates around the polar axis, which is attached to a tripod or pier. So yeah, the basic idea is exactly the same as other EQ mounts, but the mechanical layout is different,unlike a fork mount, where the scope sits between two arms, or a yoke mount, where it's inside a ring.

1.4.1 The Meridian Flip

So here's the deal with the meridian flip. Imagine you're tracking an object as it rises in the east and moves west. On a GEM, the telescope starts off on the east side of the mount. As the object crosses the

meridian (that imaginary north-south line through the zenith), the telescope eventually hits the tripod or the mount itself,it literally can't go further without crashing.

To keep tracking, you have to do a "meridian flip." You stop tracking, quickly slew the telescope to the other side of the mount (the west side), and restart tracking. The whole thing takes like 30 seconds if you're fast, but during that time, the target is obviously not being tracked.

Why does this matter? If you're doing visual observing, it's just a minor annoyance to you. You have to recenter the object after the flip. But if you're doing astrophotography and you're in the middle of a long exposure, you have to stop the exposure, do the flip, and restart a new exposure. It breaks your imaging run. Watch this YouTube video here.

Tip 4: You never know - IOAA 2017 OT

In IOAA 2017, the participants were asked to use both the Equatorial Mount Telescope in (N1) and a Dobsonian in (N2). So, it's better to learn fundamental of all even before the Telescope camp. You can just mess and play around there :D

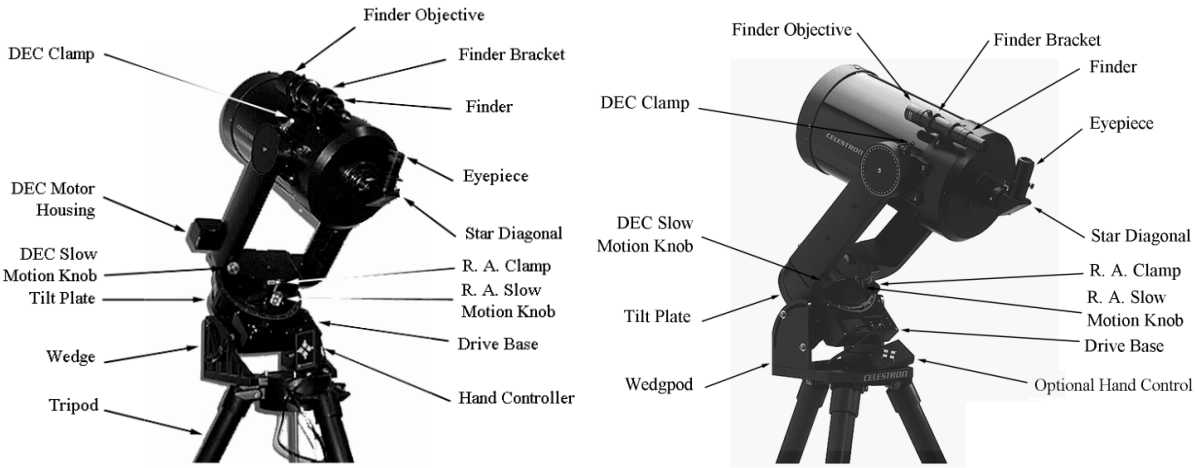
Check the IOAA 2017 Telescope round problems from here.

Task 4: IOAA 2013 - OT2

Check the question from here. This was EVIL!! They didn't allow the participants to look at the setting circle. So, practice pointing the objects in the sky without the help of setting circles too :(

1.5 Fork Mount

Fork mount is basically the other common type of equatorial mount you'll see. Instead of that T-shaped German equatorial setup, the fork mount looks like,well, a fork. Two arms come up from a base and hold the telescope tube between them. The whole fork rotates left and right for RA, and the telescope pivots up and down between the arms for Declination.



If you just put a fork mount on a flat tripod, it's actually an altazimuth mount. The base rotates left/right (azimuth) and the scope tilts up/down (altitude). To make it equatorial, you have to put it on a wedge. The wedge is a metal plate that tilts the entire fork so that the base's rotation axis points at the celestial pole. Once you add the wedge and polar align, boom! it's an equatorial mount. If you forget the wedge or set it to the wrong latitude, your tracking will drift in Dec just like any misaligned EQ mount. So

yeah, it's literally an alt-az mount tilted sideways. Once it's tilted and polar aligned, it tracks just like any other EQ mount, thus you only need to move the RA axis at the sidereal rate.

Fork mounts are super common on Schmidt-Cassegrains and Maksutovs. What's cool is that there's no counterweight needed, because the scope is centered between the arms. That makes the whole thing more compact and stable. The tube doesn't stick out to one side like on a GEM, so it's less likely to wobble. But they can get heavy and aren't as easy to balance if you add a bunch of accessories to the back.

Tip 5: Fine Adjustment on Fork Mounts

Fine adjustment knobs on fork equatorial mounts can feel a bit weird if you're used to German equatorials. On a fork mount, the fine adjustment for RA is usually on the base, near where the fork rotates, and the Dec fine adjustment is on one of the arms. The trick is: always check which axis you're actually moving before you start turning. Sometimes the knobs are close together, and in the stress of the round, it's easy to crank the Dec knob thinking you're tracking in RA, and then you watch your target slowly drift vertically out of frame.

There aren't many videos in YouTube on Fork mount. Well, there's one, and that's how to balance it (here). You don't have to worry much about it as Fork Mount isn't quite asked in IOAA.

1.6 Horseshoe Mount

The horseshoe mount is the big-dad version of a fork mount that you see on huge professional telescopes like the Hale Telescope at Palomar. It's called a horseshoe because instead of a full fork becuase it has a massive, curved, horseshoe-shaped bearing that holds the telescope at the top. The horseshoe handles all the weight of the telescope, so it's super stable. Also, these mounts are almost always computer-controlled because they're on research scopes, but in principle, you could move them manually with motors disengaged, but it would just be really slow.

2 Accessories

Okay, so you've got the telescope and the mount figured out. But honestly, the real headaches during an olympiad often come from the accessories. These are all the little extra pieces you attach to your scope, like the finderscope, eyepieces, focuser, Barlow lens, and filters, etc. They're not just add-ons; they're what actually let you see stuff, find stuff, and tweak what you're seeing. In IOAA, you'll be tested on how well you know these parts,how to use them quickly, how to choose the right one for the job, and how to fix things when they go wrong mid-task.

2.1 Finderscope

The finderscope is literally your telescope's GPS,it's that little mini-telescope or red-dot thing mounted on top of your main tube. Its whole job is to help you find stuff without having to squint through the main eyepiece at a tiny patch of sky. The main telescope has a narrow field of view, like looking through a straw, so pointing it blindly at the sky is pretty much hopeless. The finder gives you a much wider view, so you can star-hop to your target. There are two main types of finderscopes:



An Optical Finder



A Reflex Finder

1. **Optical Finders**
   Optical finders are basically mini telescopes mounted on top of your main scope. They usually come in sizes like 6x30, 8x50, or 9x60,where the first number is the magnification and the second is the aperture in millimeters. A 6x30 has 6x magnification and a 30mm front lens; an 8x50 is brighter and shows more detail because of the bigger aperture. They have crosshairs (reticles) inside that you align with your main telescope's view.

   Because they magnify, optical finders let you see fainter stars than you can with your naked eye. This makes star-hopping way easier. The downside is that they invert the image (usually upside-down and mirrored), which can be confusing if you're trying to match what you see to a star chart. Also, at high magnification, the field of view is smaller, and if your finder isn't well aligned, you'll miss the target completely.

2. **Reflex Finder**
   Reflex finders don't magnify anything. They have a clear glass window and project a lit reticle (usually

a red dot or a circle) that seems to float in the sky. You look through it with both eyes open, so you see the dot overlaid on the actual night sky. They're super intuitive because there's no image inversion, and they have a huge field of view, like around 10 degrees or more, so pointing is fast.

But because there's no magnification, you can't see fainter stars through them. If you're looking for something dim, you have to aim using bright stars you can already see with your naked eye. Parallax can be an issue too,if your eye isn't centered, the dot appears to shift relative to the stars.

2.1.1 Aligning the Finderscope

Aligning the finder is step one, and if you skip it, you're gonna have a bad time. So, aligning your finderscope is non-negotiable. If it's off, you'll point the main telescope at the empty sky every time. Here's how to do it right, whether you're in daylight or already at night. Here's the step to aligning the finderscope with the eyepiece:

- **Choose a distant target**
  You want to do this during the day using a fixed land object. At night the things in the sky are moving which makes it more difficult to get the alignment correct. During the day, set your telescope so that you can see a distant object, at least 1/4 mile away and farther is better. That gives a very well defined target right where the cross arm meets the pole. But the top of a chimney, a letter on a distant sign, or something similar can work as well. This is your alignment target.
- **Center the target in the main telescope**
  Put in a low- or medium-power eyepiece (like a 25mm) and point the telescope at your target. Get it perfectly centered in the eyepiece. Use the telescope's slow-motion controls if available, or gently nudge the tube. If it's a star, defocus it slightly so it looks like a big blob,this makes centering way easier.
- **Look through the finder & Adjust**
  Without moving the main telescope, check the finder. You'll likely see the target way off-center, maybe not even in the field of view.

  Finders have either three or six small screws around the mounting bracket. Usually, three locking screws (to hold it tight) and three adjustment screws (to push the finder around). Loosen the locking screws just enough so the finder can move, but not so much that it flops around. Then gently turn the adjustment screws to move the reticle toward the target. Move it until the target is exactly on the crosshairs.
- **Recheck and fine-tune**
  Go back to the main eyepiece. Is the target still centered? If not, recenter it in the telescope first, then check the finder again and tweak further. This might take a couple of back-and-forths. Once it's aligned, tighten the locking screws evenly. To be sure, point at another object in a different part of the sky. Center it in the main scope and verify it's also centered in the finder.

Tip 6: Finderscope in an IOAA Round

In an IOAA round, you might not get time to align the finder from scratch. So, they often give you a telescope that's already roughly aligned. You should never just assume it's perfect. Take literally 30 seconds at the start to verify: point the main scope at the brightest star you can see, center it in a medium-power eyepiece, and then check the finder. If the star is dead-center on the crosshairs, great. If it's even a little off, adjust it right then. Those few seconds will save you from chasing

nothing later.

When the task is to find something faint like a galaxy, nebula, or a tight double star, you can't just swing the finder around hoping to stumble on it. Open your star chart (they usually provide one like in the IOAA 2019, 2025) and identify a bright, easy-to-find star near your target. Point the finder at that star first. Then, using the pattern of fainter stars between that bright star and your target, move slowly, matching what you see in the finder to the chart.

Task 5: IOAA 2019 – Finderscope Alignment Breakdown

Check the Task 1 from here

Okay, so this is literally the first thing they hit you with in the 2019 telescope round. They give you a 150/750 Newtonian (that's 150mm aperture, 750mm focal length). You've got three eyepieces (25mm, 10mm) and a 2x Barlow, but for this task, you must use the 25mm. The telescope is already polar-aligned, so you don't need to touch the mount, just the finder. The task says: "The finderscope is NOT aligned at the beginning. Point the telescope at Saturn and align the finderscope parallel to the main tube." They give you 5 minutes. Sounds simple, but here's what's really going on.

First, you need to actually find Saturn in the main telescope using only the main tube, because the finder is off. Since the mount is polar-aligned and tracking, you can't just swing it around freely. You'll have to use the slow-motion controls (probably the RA and Dec fine-adjust knobs) to sweep the sky until Saturn appears in your 25mm eyepiece. Saturn is bright and distinctive (rings!), so once you see it, center it carefully. Then, without moving the main scope, you look through the finder. Saturn will probably be nowhere near the crosshairs. You now loosen the finder's adjustment screws and move the finder until Saturn is centered exactly on the crosshairs. Tighten everything down, recheck in the main scope, and if it's still centered, you're done.

But here's the catch: They say, "If the alignment of Saturn is not within the crosshair of the finderscope, the telescope assistant will correct it,and you receive only partial or no points." That means if you mess up and hand it over unaligned, they'll fix it for you, but you lose points. So precision matters.

2.2 Eyepiece

Eyepieces are the part you actually look through, they're like the lens of your telescope, but removable so you can swap them out for different magnifications. Think of them as different camera lenses: wide-angle, telephoto, etc. They take the focused light cone from your telescope and magnify it into the image you see.

Every eyepiece has a focal length, measured in millimeters. That's the number printed on the side,like 25mm, 10mm, 6mm, etc. Shorter focal length = higher magnification. The magnification formula is super simple:

$$\text{Magnification} = \frac{\text{Telescope Focal Length}}{\text{Eyepiece Focal Length}} = \frac{f_o}{f_e}$$

So if your telescope has a 750mm focal length (like the IOAA 2019 Newtonian) and you use a 25mm eyepiece, you get:

$$\frac{750}{25} = 30\times \text{ magnification}$$

Switch to a 10mm, and you're at $750/10 = 75\times$.



But magnification isn't everything. Eyepieces also have an apparent field of view (AFOV), measured in degrees. This is how wide the view feels when you look into it. Cheap eyepieces might have a narrow 40°–50° AFOV, like looking through a tunnel. Wide-field eyepieces can go 70°, 82°, or even 100°, which feels more like a window. A wider field makes finding and tracking objects easier, especially at high power.

$$\text{AFOV} = 2\tan^{-1}(\frac{d_{FS}}{2f_e})$$

This is the general formula for AFOV of the Eyepiece where d_{FS} is the field stop diameter (exit eye pupil) and f_e is the focal length of the eyepiece. I recommend you to read Instrumentation - 1 (Field of View)

Task 6: IOAA 2025 – Telescope Task (Eyepiece thing) Breakdown

Check OT01.3 from here

Alright, so this is a classic indoor screen-based telescope task from IOAA 2025. Your job is to find object 'Z' in that projected sky using your telescope, center it in the eyepiece's crosshair, and show it to the examiner.

- **Map Study First**
  Before you reach your station, you would be provided 20-30 minutes of the question reading time and would be asked to solve the above part of the question. You already have the printed map (Map-OT01). Mark object 'Z' clearly on it and memorize the star pattern around it. The screen projection is different in scale from the real sky, so you need to mentally match the pattern, not the size.
- **Start with High Focal length (Low Power) Eyepiece**
  Use the 25mm eyepiece first. It gives a wider field of view, so it's easier to find the general area and recognize the star pattern from your map. Once you see the pattern on the screen through your scope, slowly move until 'Z' is somewhere in the field. Don't worry about centering yet. Just get it in view.
- **Switch to Low Focal Length (High Power) with Crosshair**
  Then, swap to the 10mm eyepiece (which has a crosshair). This gives higher magnification and makes centering precise. Now, using fine-adjustment knobs, nudge the telescope until 'Z' is exactly on the intersection of the crosshair.
- **Scoring Breakdown**
  The scoring was super specific:

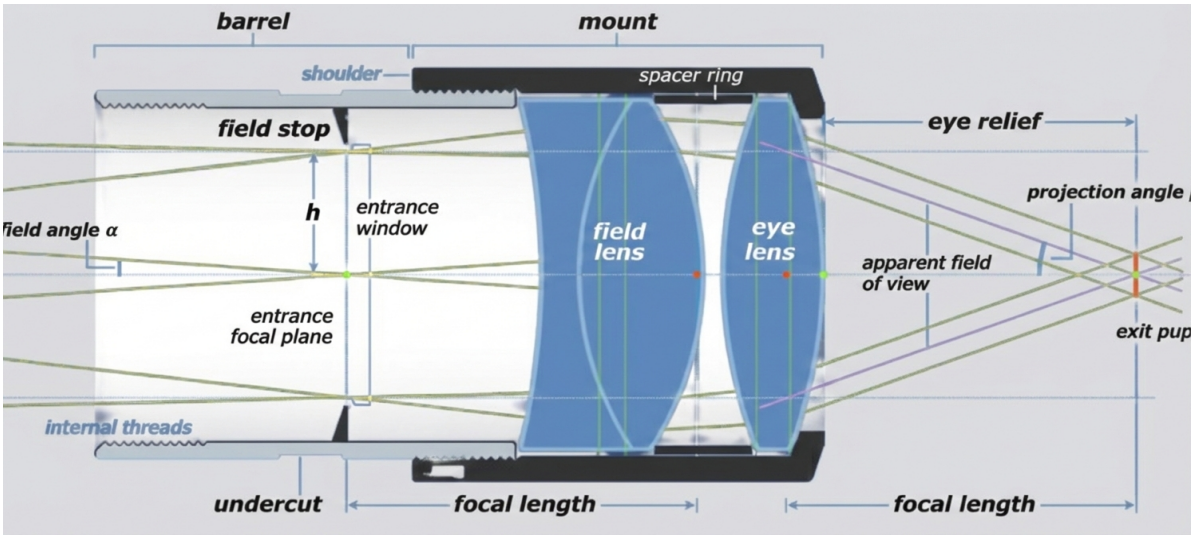
  If you use the **10mm eyepiece**:
  - 2 points for correct eyepiece choice.
  - 3 points if the object is anywhere in the field of view.
  - 6 points if any part of the object is touching the origin (center) of the crosshair.

  If you only use the **25mm eyepiece**:
  - You only get points if the object is in the field. No crosshair centering points.

In this type of observational task, try until the last second to find the object. And, until the final second, even if you fail to find the object with a low-power (25mm) eyepiece, just replace that with the high-power (10mm) eyepiece. You could get an extra 2 points just by switching the eyepieces.

So, the key is to use the 25mm for finding and the 10mm for measuring.

2.2.1 Eyepiece Internal Ray Diagram



Starting from the left side, where it says barrel, that's basically the "front door" of the eyepiece. As the light enters from your telescope, it hits the entrance focal plane. This is the exact spot where the telescope's objective lens or mirror has already done the hard work of creating a tiny, inverted version of whatever star or planet you're looking at. Right there, you've got the field stop, which is one of the most important parts. It's essentially a physical "crop tool", a ring that blocks out the messy, distorted light at the edges so you get a crisp, circular view instead of a blurry mess around the rim. It defines your field angle α, which is basically how much of the sky you're seeing at once.

Once the light passes that entrance window, it hits the glass. The first glass is the field lens. Its main job is to catch the diverging light rays and bend them back toward the center so they don't just get lost inside the mount. Then, those rays pass through the eye lens, which is the one you actually look into. This lens collimates (a fancy way of saying it makes the rays parallel again), so your eye can relax and focus on them as if they were coming from infinitely far away. The space between these lenses, held steady by that spacer ring, is what determines the overall focal length and how much magnification you're getting.

Finally, we get to the part where you actually put your face: the eye relief and exit pupil. The exit pupil is that tiny, concentrated circle of light on the far right where all those rays cross over. If you want to see the full apparent field of view, your eye's pupil has to be hovering exactly at that spot. The distance you have to hold your eye back from the lens to see clearly is the eye relief. If that distance is too short, you're basically smashing your eyelashes against the glass just to see the whole image; if it's long, it's a much more chill experience. It's all about finding that perfect "sweet spot" where the light being projected matches up perfectly with your own eye.

Let's talk about some important terms in the next page.

2.2.2 Field Stop Diameter

Alright, so you know how every eyepiece has a certain field of view? The thing that actually determines that is the field stop. It's a physical ring or diaphragm inside the eyepiece that cuts off the light cone. Think of it like a window inside the eyepiece barrel: only light that passes through this window gets to your eye. Anything outside it is blocked.

$$\tan(\text{TFOV}) = \frac{\text{Field Stop (mm)}}{\text{Telescope Focal Length (mm)}} = \frac{d_{FS}}{f_o}$$

$$\tan(\text{AFOV}) = \frac{\text{Field Stop (mm)}}{\text{Eyepiece Focal Length (mm)}} = \frac{d_{FS}}{f_e}$$

2.2.3 Exit Pupil

It's the diameter of the beam of light exiting the eyepiece, measured in millimeters. It's the bright circle you see floating above the eyepiece. You can calculate it easily:

$$\text{Exit Pupil} = \frac{\text{Telescope Aperture}}{\text{Magnification}} = \frac{D}{M}$$

Or, since magnification $= f_o/f_e$, it's also:

$$\text{Exit Pupil} = \frac{D}{f_o/f_e} = \frac{f_e}{f_o/D} = \frac{\text{Eyepiece Focal Length}}{\text{Telescope f/\#}}$$

So if you have a 150mm telescope and you're using a 25mm eyepiece at 30× magnification:

$$\text{Exit Pupil} = \frac{150}{30} = 5 \text{ mm}$$

Why does this matter? Your eye's pupil only opens about 7mm max in total darkness for a young person, less if there's light pollution or you're older. If the exit pupil is bigger than your eye's pupil, you're wasting light. The extra light just hits your iris. If it's too small (like below 1mm), the view gets dim and starts showing floaters in your eye. For deep-sky viewing, you want an exit pupil around 2–7mm. For planets, you can go smaller (1–2mm) because you're looking for detail, not brightness.

2.2.4 Eye Relief

Eye relief is the distance between the eyepiece lens and your eye where you can still see the whole field of view. It's measured in millimeters. Long eye relief (15–20mm) means you can hold your eye back comfortably, maybe even with glasses on. Short eye relief (less than 10mm) means you have to press your eyeball almost against the glass.

If you wear glasses (especially if you have astigmatism), you need long eye relief so your glasses don't scratch the lens and you can still see the full field. Even if you don't wear glasses, short eye relief is annoying. Your eyelashes smear the lens, you get kidney-beaning (where part of the field blacks out if you move slightly), and it's just uncomfortable for long sessions.

Some good eyepieces have adjustable eyecups that roll up or down to help you position your eye just right. In an olympiad setting, you usually can't choose your eyepiece, but if you have options, pick one

with comfortable eye relief so you're not squinting and losing time.

Problem 1: IOAA 2018 Telescope Round – Problems 1

Check the problem from here. The telescope had focal length $f_o = 700$mm and the focal length of the eyepiece was $f_e = 26$mm. They gave you the eyepiece's apparent field of view (AFOV) (45°). Now, Estimate the TFoV.

Solution:

They gave you the eyepiece's apparent field of view (AFOV) (45°), telescope focal length $f_o = 700$ mm, and eyepiece focal length $f_e = 26$ mm. You had to find the true field of view (TFOV). The formula they expect you to use is:

$$F \cdot \tan\left(\frac{\text{TFOV}}{2}\right) = f \cdot \tan\left(\frac{\text{AFOV}}{2}\right)$$

Because the eyepiece's field stop diameter is $D = 2f_e \tan(\text{AFOV}/2)$. That same diameter projected at the telescope's focal plane gives $D = 2f_o \tan(\text{TFOV}/2)$. Set them equal and solve. (We've discussed these formulas above and also in Instrumentation - 1, Plug in $f_o = 700$, $f_e = 26$, AFOV $= 45°$,

$$\text{TFOV} \approx 1.76° \quad \text{(round to } 1.8°\text{)}$$

Scoring was strict, 10 points for 1.6°–1.9°, 5 points for 1.3°–1.5° or 2.0°–2.2°. So if you messed up the formula or calculation by even 0.2°, you lost half the points.

Task 7: IOAA 2018 Telescope Round – Problems 2

Check the problem from here. The telescope had focal length $f_o = 700$mm and the focal length of the eyepiece was $f_e = 26$mm.

This one was purely observational. They displayed text on a screen somewhere in the room, and you had to point the telescope at it, focus, and read what it said. The text listed: two planets (Saturn, Jupiter), two Messier objects (M15, M74), and one equatorial coordinate (+30° 21h). You wrote these down for points,8 points each.

But the catch is that you couldn't just look with your eyes. You had to use the telescope. That means:

- First, you had to find the screen in your finderscope, and also, they tested mount handling: could you smoothly point the scope from one part of the room to another? If it were an equatorial mount, you might need to loosen clutches and move it manually since you're not tracking the sky.
- Then, center it in the main telescope using a low-power eyepiece (probably the 26mm).
- Possibly switch to a higher-power eyepiece if the text was small.

Idea 2: Calculating FOV of a telescope - IOAA 2022 Telescope round (Problem 2)

Check the problem from here. So, in this problem you have to calculate the TFOV of the telescope. But this time you dont have any other information. YOU JUST HAVE YOUR TELESCOPE TUBE, ALT-AZ MOUNT and YOUR EYEPIECE. Let's learn how to calculate the TFOV with this setup only.

Let's start with the definition of TFOV itself. It's basically how much of the sky (or the ground)

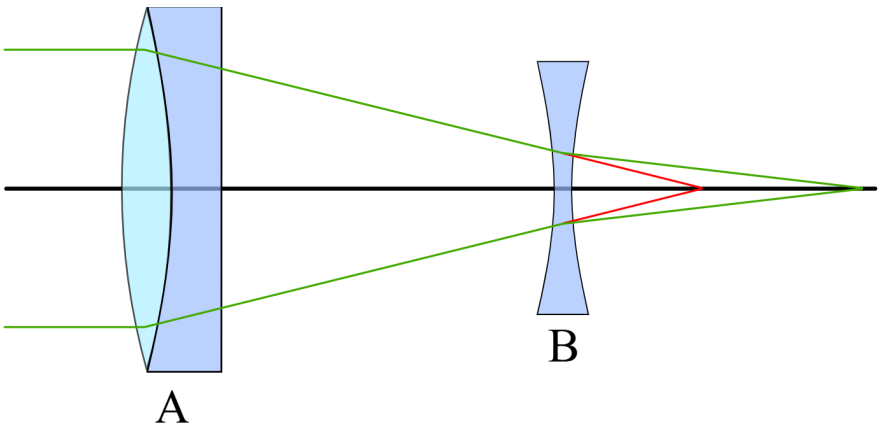
you can see at once through the eyepiece. The steps you do it is:

1. **Pick the reference object**
   Point your telescope at something easy to recognize here on Earth, like a building with clear features (windows, edges, etc.). You want something that won't move, so you can measure accurately.
2. **Move the Azimuth Knob**
   Look through your telescope and pick a starting point on the building (maybe the left edge of a window). Now, without moving the telescope up or down, just turn it slowly sideways (using one axis, like the azimuth slow motion knob) until that starting point moves out of view and another landmark appears at the opposite side of your view. Before doing this, note the initial azimuth reading in the dial (α_1)
3. **Repeat!**
   Keep doing that. Count how many entire fields of view (n) you move past until you've gone from one end of the building to the other (or between two clear landmarks). Let's say you moved n full "frames" or fields of view. After completing n frames, check the dial and note the final azimuth reading in the dial (a_2). Thus $\Delta\alpha$ (change in azimuth) is simply ($\Delta\alpha = \alpha_2 - \alpha_1$)
4. **Formulate**
   The math is super simple
   $$\text{TFOV} = \frac{|\Delta\alpha|}{n}$$
   So if your telescope moved through 6° total ($\Delta\alpha = 6°$) and you counted 5 fields of view ($n = 5$), then:
   $$\text{TFOV} = \frac{6}{5} = 1.2°$$
   That means each field of view is 1.2° wide.

This was the exact idea behind IOAA 2022 (OT2). You chose a reference point, then do the smart work as above. And then after this, we had to calculate AFOV similar to IOAA 2018 OT1 (above discussed Problem 1).

2.3 Barlow Lens

It's a tube with a lens inside that you put between your telescope and your eyepiece, and it multiplies your magnification. If you have a 2x Barlow, it doubles the power of whatever eyepiece you use. So your 25mm eyepiece suddenly acts like a 12.5mm, and your 10mm acts like a 5mm. It's way cheaper than buying a whole set of short focal length eyepieces. Watch this YouTube video on Barlow from here.



2.3.1 How Barlow Works

A Barlow is a diverging lens. It spreads out the light cone coming from your telescope before it reaches the eyepiece, which makes the eyepiece "think" the telescope has a longer focal length. The magnification factor is usually written on the side: 2x, 3x, sometimes 2.5x. The formula to calculate the effective magnification simple:

$$\text{New Effective Magnification } (M_{effective}) = \text{Original Magnification } (M_{original}) \times \text{Barlow Factor } (BF)$$

Or, in focal length terms:

$$\text{Effective Focal Length } (F_{effective}) = \frac{\text{Eyepiece Focal Length } (F_{original})}{\text{Barlow Factor } (BF)}$$

So yeah, a 2x Barlow turns a 25mm eyepiece into a 12.5mm one. But remember, it also changes the effective focal ratio of your system. If your telescope is f/5, adding a 2x Barlow makes it act like f/10 for that eyepiece. I've covered IOAA 2016 (T7) (Which was on Barlow) in Instrumentation - 1 (Page 32-34). Check it out!

Task 8: IOAA 2019 - Barlow in IOAA

Check the Task 2 from here. This is IOAA 2019 Telescope Round Questions.

So in IOAA 2019, Task 2 was all about Saturn. You had to mark Titan's position, draw the rings to scale, and measure Titan's angular distance and position angle. They gave you a 10mm eyepiece, a 25mm, and a 2x Barlow. The note said: "For task 2, we recommend using 10 mm eyepiece and Barlow 2x." That's a huge hint! They want you at high magnification so you can see Titan clearly and measure the ring dimensions accurately.

Why? Because Saturn's rings are only about 43″ across. At 30× (with the 25mm), Titan might look like a tiny dot close to the planet, and the rings will be small. But with the 10mm + 2x Barlow, you're at 150× (750mm scope / (10mm/2) = 150×). That makes Saturn bigger in the field, so you can clearly separate Titan from the planet's glare, see the ring orientation and width more precisely, and make more accurate measurements for angular distance. But the catch is, high magnification also magnifies atmospheric turbulence. If the seeing is bad, the image might be shaky, making drawing harder. Still, in 2019 they recommended it, so they expected decent conditions.

2.4 Focuser

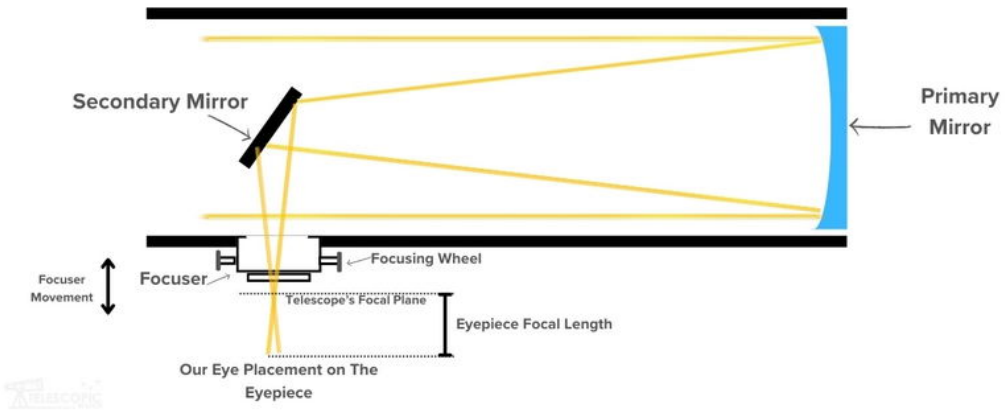
Focus in a telescope is achieved when light rays from a distant object, such as a star or planet, are converged (brought together) at a single point by the telescope's lens or mirror. When this convergence point coincides perfectly with the focal plane of our eyepiece or camera sensor, the object appears sharp and clear. Focusers help with this by sliding the eyepiece back and forth, effectively moving the focal plane of the eyepiece along with it.

Tip 7: Setup the Focuser early

Before you start your Telescope work in IOAA, adjust your focuser. If you don't do it, then you won't see anything from the eyepiece. So, first, start with a reference object. If you're indoors pointing at a screen, use text or a sharp pattern or if you're outdoor, use a bright star or planet.

Turn the focuser all the way in (eyepiece closest to the telescope), then slowly turn it out until the image snaps into focus. If you go past it, backtrack slowly. At high power, the focus point is really narrow and sensitive, like a millimeter of movement can ruin it.

In tasks where you switch eyepieces, you'll likely need to refocus each time. Different eyepieces have different optical positions, so when you swap, the focus changes. And if you add a Barlow lens, you'll need to focus again, usually moving the focuser outward.



2.5 Filters

Filters are essential tools in observational astronomy that allow you to see specific details by controlling which colors of light reach your eye or camera. Think of them like specialized sunglasses for your telescope. Each filter is designed to highlight certain features while suppressing others. The filters screw into a standard eyepiece barrel. The contrast differences are subtle, and some observers report little or no observable differences. It may take patience and experience to recognize the differences.

2.5.1 Broadband Filters

Broadband filters are the most common type of visual filter. They allow a wide range of wavelengths to pass through, typically covering a large portion of the visible spectrum. Their main purposes are to reduce glare, enhance contrast for specific colors, and protect your eyes from excessively bright objects like the Moon.

1. **Luminance Filters**
   A luminance filter, often called a neutral density (ND) filter, reduces the overall brightness of an image without changing its color balance. It works like sunglasses for your telescope, cutting down the intensity of light evenly across the visible spectrum. These are primarily used for observing very bright objects.

   The most common application is lunar observation (Provided in IOAA 2016). Without a filter, the full Moon can be painfully bright through a telescope, causing eye fatigue and reducing the visibility of subtle surface details. A typical moon filter transmits only 13%, 18%, or 25% of the incoming light. The transmission percentage is usually marked on the filter's rim. For example, an ND25 filter allows 25% of light to pass, cutting 75% of the brightness.

2. **Planetary Color Filters**
   These are colored glass filters that transmit specific color ranges while blocking others. They are identified by Wratten numbers, a standard system developed by Kodak. Each color enhances certain planetary features by increasing contrast.

   - **Red (Wratten #25)**: Transmits red and orange light while blocking blue and green. Useful for increasing contrast on Martian surface features and dust storms.
   - **Green (Wratten #58)**: Transmits green and yellow-green light. Excellent for enhancing the cloud structures and the Great Red Spot on Jupiter, and for observing Venus's atmospheric features.
   - **Blue (Wratten #80A)**: Transmits blue and violet light. Helps to see details in the cloud tops of Venus and Mars.
   - **Yellow (Wratten #12)**: Transmits yellow and some green light. Useful for improving contrast on Martian surface features and reducing glare on Venus and Jupiter.

In IOAA, you may be asked to select the appropriate filter to sketch a specific planetary detail or to estimate the visibility of a lunar feature under filtered conditions.

2.5.2 Narrowband Filters

Jarrowband filters are the next level up from those colored planetary filters. They're not just letting through "red" or "blue" light. They're designed to isolate super-specific colors of light that come from glowing gas in space. Think of them like a super precise bouncer at a club, only light with one exact wavelength gets in, everything else gets bounced.

What They Actually Isolate:
Space isn't just full of white starlight. Nebulae and certain stars emit light at very specific wavelengths called emission lines. This happens when atoms get excited and then drop down energy levels, emitting a photon with a precise energy (and therefore color). The most important ones for visual astronomy are:

- **Hydrogen-alpha (H-α):** Wavelength = 656.28 nm. This is deep red light emitted by hydrogen gas when an electron falls from the 3rd to the2nd energy level. It's what makes emission nebulae like the Orion Nebula glow.
- **Oxygen-III (O-III):** Wavelengths = 495.9 nm and 500.7 nm. These are teal-green lines from doubly ionized oxygen. They dominate in planetary nebulae (like the Ring Nebula) and supernova remnants.
- **Sulfur-II (S-II):** Wavelength = 672.4 nm. This is a deeper red line from ionized sulfur. It's often used in astrophotography to map different gas structures, but it's less common for visual use.

A narrowband filter has a super-tiny bandpass. Bandpass is the range of wavelengths it lets through. A good visual O-III filter might have a bandpass of 10-12 nm wide. An ultra-narrow one for photography can be as tight as 3 nm. The narrower it is, the more it blocks other light (like light pollution), but also the dimmer the view gets.

Why You'd Use One:
Let's say you're looking at the Orion Nebula (M42) from a suburban backyard. Without a filter, it looks like a faint gray smudge because the skyglow from streetlights washes it out. If you screw in an H-alpha or O-III filter, the filter blocks most of that white streetlight glow but lets the specific red or teal light from the nebula through. Suddenly, the nebula pops out with way more contrast and detail. It literally turns an invisible object into a visible one.

The Trade-Off:
Narrowband filters make everything darker. You're blocking like 90% of the total light. So, stars will look much fainter, often disappearing entirely unless they're emission-line stars themselves. You'll need a decently large telescope (like 6-inch or bigger) to gather enough of that specific light.

Math for Narrowband Filters:
The filter's transmission is described by its profile $T(\lambda)$. The light you see is:

$$S = \int F(\lambda) \cdot T(\lambda)\, d\lambda$$

where $F(\lambda)$ is the spectrum of the object. You need to know: if an object is strong in O-III (like the Ring Nebula), use an O-III filter. If it's an H-II region (like Orion), use H-alpha.

2.5.3 Solar Filter

Solar filters are a special and absolutely critical category of filters. Their purpose is to allow safe observation of the Sun by blocking more than 99.999% of its incoming light and harmful infrared (IR) and ultraviolet (UV) radiation. **Warning:** Using an improper filter or misusing a solar filter can cause instant and permanent eye damage, including blindness. Solar observation requires strict safety protocols.

Solar filters work by reflection and absorption. They are not like neutral density filters; they are specifically engineered to reject the immense heat and radiation of the Sun. There are two main types based on where they are placed:

- **Aperture Filters (Full-Aperture Filters):** These are placed over the front opening (aperture) of the telescope. They block the sunlight **before** it enters the optical tube. This is the only safe method for visual solar observation with most telescopes.
- **Herschel Wedges (for Refractors):** This is a specialized prism-based system used with refracting telescopes. It reflects a tiny, safe percentage of sunlight to the eyepiece while diverting the vast majority of the heat and light out of the system.

In an olympiad setting, solar observation tasks are rare due to the safety risk and scheduling, but they can occur. If presented, it will be under tightly controlled conditions.

2.5.4 Light Pollution Filters

Light pollution filters are designed to block the specific wavelengths emitted by common artificial light sources, particularly mercury vapor (at 546 nm) and high/low-pressure sodium vapor lamps (around 589 nm and 570-620 nm), while allowing the continuum and emission lines from celestial objects to pass through.

2.6 Stopwatch

In IOAA practical rounds, a stopwatch is often provided to make timing-based measurements. Here, we'll see different types of questions that have appeared in IOAA and Stopwatch. In the circular, they'll say whether you'll be provided with a stopwatch, headlight, or neither (3-4 months ahead of the IOAA itself). So with that in mind, you can guess what kind of question might be asked. I've covered some of the problems that appeared with the stopwatch.

Idea 3: IOAA 2016 – Drift Time Calculation

Check IOAA 2016 OT3 from here. In this task, you're given a telescope, a moon filter, an eyepiece with a crosshair, and a stopwatch. Your objective is to estimate the diameter of Mare Serenitatis (a large, circular lunar mare) as a fraction of the Moon's overall diameter using the drift-time method.

Setup:

First, you point the telescope at the Moon and attach the moon filter. This filter reduces the Moon's intense brightness, making it comfortable to view and preventing eye fatigue during the detailed measurement. The eyepiece has a crosshair, which is essential for precise timing as objects drift across the field of view due to Earth's rotation.

The method of calculation is based on a simple principle: the time it takes for an object to drift across the fixed crosshair in the eyepiece is proportional to its angular size. Since the telescope's field of view is constant, a larger angular object takes longer to drift past the crosshair.

Measurement Steps:

- **Align the Crosshair:** Orient the crosshair so that the Moon drifts perpendicularly across one of the lines. Typically, you align it east-west, as the Moon's drift is primarily in right ascension. (Checks your ability to point, focus, and use an eyepiece with a crosshair.)
- **Time the Full Moon's Diameter:** Place the Moon's eastern limb on the crosshair, then start the stopwatch the moment the limb touches the line. Stop the watch the moment the western limb leaves the crosshair. Record this time as t_{Moon}.
- **Time Mare Serenitatis:** Now center Mare Serenitatis in the field. Similarly, record how long it takes for this circular mare to drift completely across the same crosshair. Record this as t_{MSr}.

Calculation:

The ratio of the drift times gives the ratio of their angular diameters:

$$\frac{D_{\text{MSr}}}{D_{\text{Moon}}} = \frac{t_{\text{MSr}}}{t_{\text{Moon}}}$$

Therefore, the estimated diameter of Mare Serenitatis as a fraction of the lunar diameter is simply the ratio of the two measured times.

Idea 4: IOAA 2015 – FOV estimation by Transit Time method

We've already discussed the maths behind this in Instrumentation - 1, so before proceeding, I recommend that you guys understand that. Check IOAA 2015 (Main Telescope) Page 1, here. In Participants were first asked to point their telescope to the open cluster NGC 6475 (Messier 7) using its equatorial coordinates (RA: $17^{\text{h}}53^{\text{m}}54^{\text{s}}$, Dec: $-34°49'$). The telescope was mounted on an equatorial mount that was presumably polar-aligned. You were given a 15mm eyepiece.

For the subsequent FOV estimation task, participants needed to switch from M7 to a bright star (tho not mentioned in the instruction, but in my opinion, this is necessary because you can't just track M7 all the way, cuz its faint) at a similar declination. A suitable choice was Shaula (λ Scorpii), which is near M7 and has a comparable declination (approximately $-37°$), minimizing the need for large declination adjustments and simplifying the measurement.

Procedure for FOV Measurement:

1. Position the star at the eastern edge of the field, but on the midline. Don't just plop it anywhere, move the telescope so the star is on the *eastern* side of the field, but exactly on the horizontal line that runs through the center.
2. Wait for it to reach the center, then start the timer. Let the star drift naturally toward the center of the field. The moment it crosses the exact center (the intersection of your crosshair), that's when you hit start on the stopwatch.
3. Stop the timer when it reaches the western edge. Keep tracking it with your eyes. Don't touch the telescope. The star will keep drifting west. The moment it touches the *western* edge of the field, hit stop. That time (t_{half}) is the time it took to travel across the *radius* of the field.
4. The time you just measured is for half the field (center to edge). We need the full diameter time, just multiply t_{half} by 2.

$$t_{full} = t_{half} \times 2$$

Now just, plug your time t_{new} (in seconds) and the star's declination δ into:

$$\text{FOV [in }^\circ] = \omega \cdot \cos(\delta) \cdot t_{new} = \frac{360^\circ}{23^{\text{hr}}56^m4^s} \cdot \cos(\delta) \cdot t_{new} = \frac{t_{new} \cdot \cos(\delta) \cdot 360}{86164}$$

In IOAA 2013, too, a problem with same implication was asked. You can check it from here.

Task 9: IOAA 2025 - Lensing Time Delay

I recommend you try this by yourself. This question can be entirely done while you are sitting in your chair (except for pointing your telescope at the screen). Grab a stopwatch by yourself. here - Question (page 4-5) is the question and marking scheme. here - Simulation video is the video (start the video from 14:40), and finally here - Summary Answer Sheet is the summary sheet.

You can find everything in the IOAA 2025 Website (Problem and Solutions), here - IOAA 2025 Problem and Solutions.

3 Further Resources

1. **IOAA Past Papers**
   It is necessary to practice past papers to get familiar with the type of problems that could appear in the NAO. You can access all IOAA past papers through the provided Drive link. *It's created by AstroSandbox*
   IOAA Past Papers – Google Drive

2. **SAO (Singapore Astronomy Olympiad) Past Papers**.
   The SAO resources are a great complement to the IOAA.
   SAO Past Papers – Google Drive

   here's their website: *SAO*

3. **Star Charts 101 and Practices**
   Compiled by Fahim Rajit Hossain, the Bangladeshi IOAA team leader. They are available in the shared resource drive.
   Resource – Google Drive

   here's his GitHub repository

   Rajit13/Star-Maps-101-and-Practices

4. **AO Guide**
   It's one of the best guides for IOAA level preparation one can find online, made up by Akshat Srivastava (India, IOAA'25 Gold). He also crafted a web app to play with spherical diagrams.

   AO Guide, *Celestial Sphere*

5. **Spherical Trig and Stellar Physics Handouts**
   It's an amazingly crafted handout by Sean Ken M. Galanza (Philippines, IOAA'24 Bronze). Reading this handout will help you build a solid and strong grasp of Spherical Trig and Stellar Physics.
   Handouts - Google drive

6. **AstroSandbox**
   It was founded by some previous Ukrainian IOAA folks. AstroSandbox gives lectures, organizes events, manages clubs, composes their own astronomy tasks, as well as collects existing problems from Olympiads and public sources.
   AstroSandbox

7. **Communities**

   (a) *AODS - Astronomy Olympiad Discord Server*

   (b) *Nepal Astronomy Society Discord Server*

   (c) *Nepal Physics Olympiad Discord Server*

   (d) *Learn2Lead Nepal Discord Server*

   (e) *Olympiad Prep Hub Discord Server*

8. **Some other resources:**

   (a) *This Drive contains all the sources used in AOGuide by Akshat and more. Many national and international olympiad papers are also available here.*

(b) *AOGuide - Resources*

(c) *OAC*

(d) *OAAO*

(e) *Astronomy Olympiad Archives by Stefan Ivanov*

(f) *Resources Drive by Sudip Rokaya (NAO'23 - Absolute Winner)*

(g) *National Page Archives*